

# TERRITOIRE\_Ω OPERATIONAL REPORT

**Zone:** ZONE\_FULL\_8\_OVERLAYS\_V2

**Doctrine:** BCE-4X\_ULTIME\_ABSOLU\_ANTI\_GÉNÉRIQUE\_STRICT

**Ordre:** P15\_TERRITOIRE\_Ω\_REPORT\_CREATE\_Ω

**Generated UTC:** 2026-05-08T19:55:22+00:00

**SHA-256:** 04e67ff21bb6e4c0d7ed2e11ffa802f23652984f487af9a446fcfe01dfff757

V30 LOCK: INVOLÉ · FUSION ADD-ONLY · ANTI-GÉNÉRIQUE STRICT

## Chaîne cryptographique

**merkle\_root\_last\_sha256:**

09be873b0cea2714a06327199e3c3b206341d337071fbea641fdc41b84e888a1

**merkle\_verdict:** MERKLE\_TREE\_ANCHOR\_HOOK\_ACTIVATED

**multi\_signature\_status:** MULTI\_SIGNATURE\_VERIFICATION\_HOOK\_ACTIVATED

**ots\_automation\_status:** OTS\_UPGRADE\_AUTOMATION\_HOOK\_ACTIVATED

**chain\_integrity\_claim:** Ed25519 + PGP + Merkle + Bitcoin OTS (anti-générique strict)

## Overlays sources

| Overlay                | Status  | Last SHA (16)    | Updated UTC         |
|------------------------|---------|------------------|---------------------|
| habitat_complete_merge | PRESENT | —                | 2026-05-08T19:54:33 |
| merkle_tree_anchor     | PRESENT | 09be873b0cea2714 | 2026-05-08T15:23:52 |
| multi_year_dense_grid  | PRESENT | a541926b77a60776 | 2026-05-08T12:33:54 |
| multi_signature        | PRESENT | 16ae7c6832251889 | 2026-05-08T19:54:34 |
| territoire_visualizer  | PRESENT | fbd6386ebe3ad5ee | 2026-05-08T19:54:34 |
| anthropogenic_pressure | PRESENT | b7c5d9a5a90d3c81 | 2026-05-08T02:04:31 |
| temporal_rut           | PRESENT | ca6937d8d7a3c68b | 2026-05-08T02:29:28 |
| ots_upgrade_automation | PRESENT | 24d847d8016f3142 | 2026-05-08T17:08:28 |

## Recommandations (4)

**[P2] HABITAT\_INTEGRITY** — Continuer la surveillance 12/12 outputs habitat\_complete\_merge; pas de dégradation détectée.

**[P1] ANTHROPOGENIC\_PRESSURE** — Croiser pression anthropique OSM+WorldPop avec zones de repos; relocaliser affûts si p95>0.7.

**[P1] TEMPORAL\_RUT** — Activer fenêtres rut doctrinales (octobre-novembre cerf / septembre orignal).

**[P2] DECADAL\_TREND** — Analyse Mann-Kendall 10 ans active : surveiller tendances NDVI significatives (p<0.05) pour planification long-terme.

COMMANDANT STEEVE-MAX · BCE-4X ULTIME ABSOLU